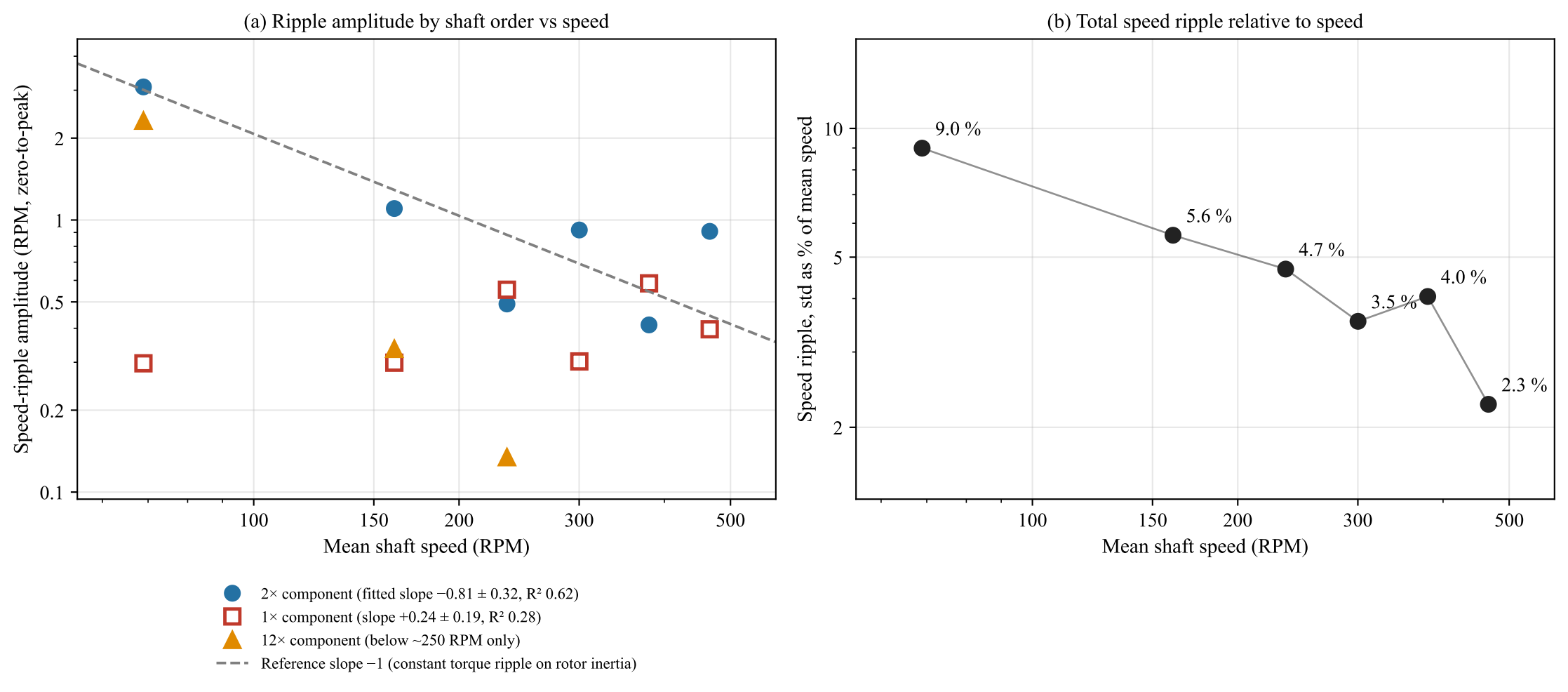
(a) Ripple amplitude by shaft order vs speed
Speed-ripple amplitude (RPM, zero-to-peak)
2
1
0.5
0.2
0.1
100
150
200
300
500
Mean shaft speed (RPM)
2× component (fitted slope −0.81 ± 0.32, R² 0.62)
1× component (slope +0.24 ± 0.19, R² 0.28)
12× component (below ~250 RPM only)
Reference slope −1 (constant torque ripple on rotor inertia)
(b) Total speed ripple relative to speed
Speed ripple, std as % of mean speed
10
5
2
9.0 %
5.6 %
4.7 %
3.5 %
4.0 %
2.3 %
100
150
200
300
500
Mean shaft speed (RPM)